

Hamilton - Jacobi Quantum Mechanics

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1 Prerequisites From Classical Mechanics

In classical Hamiltonian dynamics, the equations of motion are given by,

$$\dot{q}_i = \frac{\partial H(q, p, t)}{\partial p_i} \quad (1.1)$$

$$\dot{p}_i = -\frac{\partial H(q, p, t)}{\partial q_i} \quad (1.2)$$

If all the coordinates q_i are cyclic (hamiltonian is independent of $q_i \forall i$), and hamiltonian is a constant of motion, then the solutions for the hamilton's equations become trivial.

$$\dot{p}_i = 0 \quad \Rightarrow \quad p_i = \alpha_i = \text{const.}$$

Then the hamiltonian is only a function of α_i only. Then,

$$\dot{q}_i = \frac{\partial H}{\partial \alpha_i} = \omega_i \quad (1.3)$$

As ω_i are functions of only α_i , which are constants in time, ω_i are also constants in time. Therefore,

$$q_i = \omega_i t + \beta_i \quad (1.4)$$

This along with initial conditions of q_i and p_i solves the problem completely.

If in our problem the coordinates are not cyclic, we seek for coordinate transforms such that in the new set of coordinates, the hamiltonian is independent of as many coordinates as possible. Let the transformation be,

$$\begin{aligned} q_i &\rightarrow Q_i(q_j, p_j, t) \\ p_i &\rightarrow P_i(q_j, p_j, t) \end{aligned} \quad (1.5)$$

And let $K(Q_i, P_i, t)$ be the new hamiltonian.

But for this coordinate transform to be beneficial, we require the hamilton's equations to retain their form. Which means,

$$\dot{Q}_i = \frac{\partial K(Q, P, t)}{\partial P_i} \quad (1.6)$$

$$\dot{P}_i = -\frac{\partial K(Q, P, t)}{\partial Q_i} \quad (1.7)$$

Both the hamiltonians must obey the Least action principle. Hence,

$$\delta \int_{t_1}^{t_2} dt (p_i \dot{q}_i - H) = 0$$

and,

$$\delta \int_{t_1}^{t_2} dt (P_i \dot{Q}_i - K) = 0$$

The integrand is the lagrangian in each case.

If we add a total time derivative of a function say F that depends on any combination of q ,

p , Q or P , the variational principle will not change. This is because as the variation of these coordinates at the endpoints are zero, variation of F at the endpoints will also be zero, as we are adding a total time derivative to a time integral's integrand, the contribution of F will be only from the endpoints, which is identically zero.

Hence, We can say,

$$\lambda (p_i \dot{q}_i - H) = P_i \dot{Q}_i - K + \frac{dF}{dt}$$

We can set λ to 1 without loosing any generality as λ only corresponds to a rescaling of variables. Therefore,

$$p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{dF}{dt} \quad (1.8)$$

Such transformations of coordinates are called canonical transformations! Canonical transformations do preserve the poisson bracket relations. ie, if we had $\{q_i, p_j\} = \delta_{ij}$ to start with, then $\{Q_i, P_j\} = \delta_{ij}$ will be also true.

Now if $F = F(q, Q, t)$, then we get,

$$p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{\partial F}{\partial q_i} \dot{q}_i + \frac{\partial F}{\partial Q_i} \dot{Q}_i + \frac{\partial F}{\partial t}$$

by comparing coefficients,

$$p_i = \frac{\partial F}{\partial q_i} \quad P_i = -\frac{\partial F}{\partial Q_i} \quad K = H + \frac{\partial F}{\partial t} \quad (1.9)$$

If we take $F = \bar{F}(q, P, t) - Q_i P_i$ then, from 1.8,

$$\begin{aligned} p_i \dot{q}_i - H &= P_i \dot{Q}_i - K - P_i \dot{Q}_i - Q_i \dot{P}_i + \frac{d\bar{F}}{dt} \\ &= -K - Q_i \dot{P}_i + \frac{\partial \bar{F}}{\partial q_i} \dot{q}_i + \frac{\partial \bar{F}}{\partial P_i} \dot{P}_i + \frac{\partial \bar{F}}{\partial t} \end{aligned}$$

by comparing coefficients,

$$p_i = \frac{\partial \bar{F}}{\partial q_i} \quad Q_i = \frac{\partial \bar{F}}{\partial P_i} \quad K = H + \frac{\partial \bar{F}}{\partial t} \quad (1.10)$$

and so on...

2 Hamilton - Jacobi Theory (Classical)

If we seek a canonical transformation which takes the variables at time t to the initial state $q(0)$ and $p(0)$, then we can solve the problem by just inverting the transformations. So, we need a canonical transformation such that the new variables are constants (as they are the initial conditions, they indeed do not depend on time). Hence,

$$\begin{aligned} \frac{\partial K}{\partial P_i} &= \dot{Q}_i = 0 \\ -\frac{\partial K}{\partial Q_i} &= \dot{P}_i = 0 \end{aligned} \quad (2.1)$$

Change the notation, $S = \bar{F}$. Then we know that,

$$K = H + \frac{\partial S}{\partial t}$$

For the new variables to be constants, we need K to be a constant. Choose it to be zero. We know that, $p_i = \frac{\partial S}{\partial q_i}$ Then we have,

$$H\left(q_i, \frac{\partial S}{\partial q_i}, t\right) + \frac{\partial S}{\partial t} = 0 \quad (1 \leq i \leq n) \quad (2.2)$$

This is the Hamilton Jacobi Equation. S is called Hamilton's principle function. If we solve the equation, there will be $n+1$ constants of integration, one of which is an additive constant. As the equation only has derivatives of S , additive constant is not important. So we have n constants of integration say α_i . After solving, we have,

$$S = S(q_1, \dots, q_n; \alpha_1 \dots \alpha_n; t)$$

Choose the α_i to be the new momenta P_i . Now solve $p_i = \frac{\partial S(q, \alpha, t)}{\partial q_i}$ for $t = 0$ and find α_i in terms of the initial conditions. Now solve $Q_i = \beta_i = \frac{\partial S(q, \alpha, t)}{\partial \alpha_i}$ for initial conditions. Then invert these two equations to get $q_i = q_i(\alpha, \beta, t)$ and $p_i = p_i(\alpha, \beta, t)$!! This process, if doable, will provide us with the solution to the problem we are seeking!

If the hamiltonian is independent of time, then we can write,

$$S(q, \alpha, t) = W(q, \alpha) - Et \quad (2.3)$$

Where W is called Hamilton's Characteristic function.

$$\frac{dW}{dt} = \frac{\partial W}{\partial q_i} \dot{q}_i = \frac{\partial S}{\partial q_i} \dot{q}_i = p_i \dot{q}_i$$

Which imply,

$$W = \int p_i dq_i$$

The action variable is defined to be,

$$J = \oint_C p dq \quad (2.4)$$

Where the contour C encloses a complete period.

3 Canonical Transformations In Quantum Mechanics

Transformations in quantum mechanics are generalizations of rotations in Hilbert space. As canonical transformations preserve the poisson bracket relations in classical mechanics, we assume that quantum canonical transformations will preserve the commutation relations. Let $U(t)$ be such a transformation, and $[\hat{x}, \hat{p}] = i\hbar$ let,

$$\hat{X} = U \hat{x} U^\dagger \quad \hat{P} = U \hat{p} U^\dagger$$

$$[\hat{X}, \hat{P}] = U\hat{x}U^\dagger U\hat{p}U^\dagger - U\hat{p}U^\dagger U\hat{x}U^\dagger$$

if $U^\dagger U = UU^\dagger = \mathbb{1}$ i.e, U is unitary, then

$$\begin{aligned} [\hat{X}, \hat{P}] &= U\hat{x}pU^\dagger - U\hat{p}\hat{x}U^\dagger \\ &= U[\hat{x}, \hat{p}]U^\dagger \\ &= i\hbar UU^\dagger \\ &= i\hbar \end{aligned}$$

So, unitary transformations preserve the commutation bracket relations! So, we can assume that Canonical transformations in quantum mechanics are unitary transformations. Now, let $|x\rangle$ be the eigenvectors of $\hat{x}$ with eigenvalues x . Then,

$$\hat{x} |x\rangle = x |x\rangle \quad \Rightarrow \quad U\hat{x}U^\dagger(U |x\rangle) = x(U |x\rangle)$$

$$\Rightarrow \hat{X}(U |x\rangle) = x(U |x\rangle)$$

Hence,

$$|X\rangle = U |x\rangle \tag{3.1}$$

Similarly, $|P\rangle = U |p\rangle$. Here, as U is time dependent, $|X\rangle$ and $|P\rangle$ are time dependent. Also note that as canonical transformations are unitary, they preserve the eigen value spectrum. Now, Define,

$$\begin{aligned} \langle x_1 | X_2 \rangle &= \langle x_1 | U |x_2\rangle = e^{\frac{i}{\hbar}F_1(x_1, X_2, t)} \\ \langle x_1 | P_2 \rangle &= \langle x_1 | U |p_2\rangle = e^{\frac{i}{\hbar}F_2(x_1, P_2, t)} \end{aligned} \tag{3.2}$$

and so on.. Here F_1 and F_2 etc. are the generating function(functions of operators) of the quantum canonical transformations. From the analogy of classical mechanics, we can define,

$$\hat{p} = \frac{\partial \hat{F}_2(\hat{x}, \hat{P}, t)}{\partial \hat{x}} \quad \hat{Q} = \frac{\partial \hat{F}_2(\hat{x}, \hat{P}, t)}{\partial \hat{P}} \tag{3.3}$$

4 Quantum Hamilton-Jacobi Equation

To solve a quantum mechanical problem, we usually solve the Schrodinger equation,

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}(\hat{x}, \hat{p}, t) |\psi(t)\rangle \tag{4.1}$$

We know that,

$$|X\rangle_t = U(t) |x\rangle$$

We would like to know how the Schrodinger equation looks in the new transformed basis. So for that,

$$i\hbar \langle X | \frac{\partial}{\partial t} |\psi(t)\rangle = \langle X | H |\psi(t)\rangle$$

As $\langle X |$ is time dependent, we cannot take that inside the time derivative. But,

$$\langle X | \frac{\partial}{\partial t} |\psi\rangle = \frac{\partial}{\partial t} \langle X | \psi\rangle - \left(\frac{\partial}{\partial t} (\langle X |) \right) |\psi\rangle$$

Using this, we get,

$$i\hbar \left[\frac{\partial}{\partial t} \langle X | \psi\rangle - \left(\frac{\partial}{\partial t} (\langle X |) \right) |\psi\rangle \right] = \langle X | H |\psi\rangle$$

But $\langle X | = \langle x | U^\dagger$. Then,

$$i\hbar \left[\frac{\partial}{\partial t} \langle X | \psi\rangle - \left(\frac{\partial}{\partial t} (\langle x | U^\dagger) \right) |\psi\rangle \right] = \langle X | H |\psi\rangle$$

But $\langle x |$ do not depend on time. So we can take that out of the time derivative.

$$i\hbar \left[\frac{\partial}{\partial t} \langle X | \psi\rangle - \langle x | \frac{\partial U^\dagger}{\partial t} |\psi\rangle \right] = \langle X | H |\psi\rangle$$

Let us denote the wave function in the new basis as $\psi^X(X, t) = \langle X | \psi\rangle$ and wave function in the old basis as $\psi^x(x, t) = \langle x | \psi\rangle$. Also, note the completeness relation in the new basis,

$$\int dX' |X'\rangle \langle X'| = \mathbb{1}$$

Using these two,

$$i\hbar \left[\frac{\partial \psi^X(X, t)}{\partial t} - \int dX' \langle x | \frac{\partial U^\dagger}{\partial t} |X'\rangle \langle X'| \psi\rangle \right] = \int dX' \langle X | H |X'\rangle \langle X'| \psi\rangle$$

$$\Rightarrow i\hbar \frac{\partial \psi^X(X, t)}{\partial t} = \int dX' \langle X | H |X'\rangle \psi^X(X', t) + \int dX' \langle x | i\hbar \frac{\partial U^\dagger}{\partial t} |X'\rangle \psi^X(X', t)$$

$|X\rangle = U |x\rangle \Rightarrow \langle x | = \langle X | U$. Then,

$$i\hbar \frac{\partial \psi^X(X, t)}{\partial t} = \int dX' \langle X | \left[H + i\hbar U \frac{\partial U^\dagger}{\partial t} \right] |X'\rangle \psi^X(X', t)$$

Define, the new hamiltonian, as

$$K = H + i\hbar U \frac{\partial U^\dagger}{\partial t} \tag{4.2}$$

Then,

$$\begin{aligned} \langle X | \left[H + i\hbar U \frac{\partial U^\dagger}{\partial t} \right] |X'\rangle &= \langle X | K |X'\rangle \\ &= K \left(X, -i\hbar \frac{\partial}{\partial X}, t \right) \langle X | X'\rangle \\ &= K \left(X, -i\hbar \frac{\partial}{\partial X}, t \right) \delta(X - X') \end{aligned}$$

Then,

$$\begin{aligned} i\hbar \frac{\partial \psi^X(X, t)}{\partial t} &= \int dX' K \left(X, -i\hbar \frac{\partial}{\partial X}, t \right) \delta(X - X') \psi^X(X', t) \\ \Rightarrow i\hbar \frac{\partial \psi^X(X, t)}{\partial t} &= K \left(X, -i\hbar \frac{\partial}{\partial X}, t \right) \psi^X(X, t) \end{aligned}$$

The Schrodinger equation in the new transformed position basis looks like this. We have discovered how the hamiltonian transforms when we do the canonical transformations. It transforms like equation 4.2. Now, using the second term in equation 4.2,

$$\begin{aligned} \langle x_1 | i\hbar U \frac{\partial U^\dagger}{\partial t} | x_2 \rangle &= i\hbar \langle x_1 | \left[\frac{\partial(UU^\dagger)}{\partial t} - \frac{\partial U}{\partial t} U^\dagger \right] | x_2 \rangle \\ &= i\hbar \langle x_1 | \left[\frac{\partial(\mathbb{1})}{\partial t} - \frac{\partial U}{\partial t} U^\dagger \right] | x_2 \rangle \\ &= - \langle x_1 | i\hbar \frac{\partial U}{\partial t} U^\dagger | x_2 \rangle \\ &= - \int dx_3 \langle x_1 | i\hbar \frac{\partial U}{\partial t} | x_3 \rangle \langle x_3 | U^\dagger | x_2 \rangle \end{aligned}$$

$|x_3\rangle$ is independent of time. $\langle x_3 | U^\dagger = \langle X_3 |$ and $dx_3 = dX_3$ as canonical transformations preserve the eigenvalue spectrum. Hence,

$$\begin{aligned} \langle x_1 | i\hbar U \frac{\partial U^\dagger}{\partial t} | x_2 \rangle &= - \int dX_3 i\hbar \frac{\partial}{\partial t} \langle x_1 | X_3 \rangle \langle X_3 | x_2 \rangle \\ &= - \int dX_3 i\hbar \frac{\partial}{\partial t} \left(e^{\frac{i}{\hbar} S(x_1, X_3, t)} \right) \langle X_3 | x_2 \rangle \\ &= - \int dX_3 i\hbar \left(\frac{i}{\hbar} \right) \frac{\partial S}{\partial t} e^{\frac{i}{\hbar} S(x_1, X_3, t)} \langle X_3 | x_2 \rangle \\ &= \int dX_3 \frac{\partial S}{\partial t} \langle x_1 | X_3 \rangle \langle X_3 | x_2 \rangle \\ &= \int dX_3 \langle x_1 | \frac{\partial \hat{S}(\hat{x}, \hat{X}, t)}{\partial t} | X_3 \rangle \langle X_3 | x_2 \rangle \\ &= \langle x_1 | \frac{\partial \hat{S}}{\partial t} | x_2 \rangle \end{aligned}$$

Hence, equation 4.2 can be written as,

$$\hat{K} = \hat{H} + \frac{\partial \hat{S}}{\partial t} \quad (4.3)$$

This is exactly the equation we had in the classical case, but in an operator form!! Now, just like in the classical case, we need the new transformed hamiltonian to be zero. Setting $K = 0$ gives us the Quantum Hamilton Jacobi equation,

$$H \left(\hat{x}, \frac{\partial \hat{S}(\hat{x}, \hat{X}, t)}{\partial \hat{x}}, t \right) + \frac{\partial \hat{S}(\hat{x}, \hat{X}, t)}{\partial t} = 0 \quad (4.4)$$

This operator differential equation solved will give us $\hat{S}$.

$\hat{S}$ is the Quantum version of Hamilton's principle function. With this, the wave function in the old variables can be found easily,

$$\begin{aligned}\psi^x(x_1, t) &= \langle x_1 | \psi(t) \rangle \\ &= \int dX_2 \langle x_1 | X_2 \rangle \langle X_2 | \psi(t) \rangle \\ &= \int dX_2 e^{\frac{i}{\hbar} S(x_1, X_2, t)} \psi^X(X_1, t)\end{aligned}\tag{4.5}$$

The Quantum version of H-J equation we now have is an operator differential equation. We want a C-number differential equation because operator differential equations are difficult to handle. For that, let $U(t) = T(t)A$. The time dependence of $U(t)$ is completely captured by $T(t)$ and A is time independent. Both A and $T(t)$ are unitary. Then,

$$e^{\frac{i}{\hbar} S} = \langle x_1 | T(t)A | x_2 \rangle$$

Differentiate with respect to t,

$$\frac{i}{\hbar} \frac{\partial S}{\partial t} e^{\frac{i}{\hbar} S} = \langle x_1 | \frac{\partial T(t)}{\partial t} A | x_2 \rangle$$

From Quantum HJ equation,

$$\begin{aligned}0 &= H + \frac{\partial \hat{S}}{\partial t} \\ &= H + i\hbar U \frac{\partial U^\dagger}{\partial t} \\ &= H + i\hbar T A A^\dagger \frac{\partial T^\dagger}{\partial t} \\ &= H + i\hbar T \frac{\partial T^\dagger}{\partial t}\end{aligned}$$

As $TT^\dagger = \mathbb{1}$, $T \frac{\partial T^\dagger}{\partial t} = -T^\dagger \frac{\partial T}{\partial t}$. Hence,

$$H = i\hbar T^\dagger \frac{\partial T}{\partial t} \quad \Rightarrow \quad \frac{\partial T}{\partial t} = \frac{1}{i\hbar} HT$$

Therefore,

$$\begin{aligned}\frac{i}{\hbar} \frac{\partial S}{\partial t} e^{\frac{i}{\hbar} S} &= \frac{1}{i\hbar} \langle x_1 | HTA | x_2 \rangle \\ &= \frac{1}{i\hbar} \langle x_1 | HU | x_2 \rangle \\ &= \frac{1}{i\hbar} \langle x_1 | H | X_2 \rangle \\ &= \frac{1}{i\hbar} \hat{H} \left(x_1, -i\hbar \frac{\partial}{\partial x_1}, t \right) \langle x_1 | x_2 \rangle \\ &= \frac{1}{i\hbar} \hat{H} \left(x_1, -i\hbar \frac{\partial}{\partial x_1}, t \right) e^{\frac{i}{\hbar} S}\end{aligned}$$

Hence we have the C-number version of H-J equation as,

$$\left[\hat{H} \left(x_1, -i\hbar \frac{\partial}{\partial x_1}, t \right) + \frac{\partial S}{\partial t} \right] e^{\frac{i}{\hbar} S} = 0 \quad (4.6)$$

If the Hamiltonian is of the form,

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}, t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t) \quad (4.7)$$

$$\begin{aligned} \left[\hat{H} \left(x_1, -i\hbar \frac{\partial}{\partial x_1}, t \right) + \frac{\partial S}{\partial t} \right] e^{\frac{i}{\hbar} S} &= \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t) + \frac{\partial S}{\partial t} \right] e^{\frac{i}{\hbar} S} \\ &= -\frac{\hbar^2}{2m} \frac{\partial}{\partial x} \left[\frac{i}{\hbar} \frac{\partial S}{\partial x} e^{\frac{i}{\hbar} S} \right] + V e^{\frac{i}{\hbar} S} + \frac{\partial S}{\partial t} e^{\frac{i}{\hbar} S} \\ &= -\frac{\hbar i}{2m} \left[\frac{i}{\hbar} \left(\frac{\partial S}{\partial x} \right)^2 e^{\frac{i}{\hbar} S} + \frac{\partial^2 S}{\partial x^2} e^{\frac{i}{\hbar} S} \right] + V e^{\frac{i}{\hbar} S} + \frac{\partial S}{\partial t} e^{\frac{i}{\hbar} S} \end{aligned}$$

Therefore using 4.6,

$$\frac{1}{2m} \left(\frac{\partial S}{\partial x} \right)^2 - \frac{\hbar i}{2m} \frac{\partial^2 S}{\partial x^2} + V(x, t) + \frac{\partial S}{\partial t} = 0 \quad (4.8)$$

This is the C-number version of Quantum HJ equation. Note that as $\hbar \rightarrow 0$, the equation reduces to the classical HJ equation for the same kind of hamiltonian. Now, as in the classical case, if Hamiltonian is independent of time, we can write,

$$S = W - Et \quad (4.9)$$

Where W is the quantum analogue of the Characteristic function. In this case, we get,

$$\frac{1}{2m} \left(\frac{\partial W}{\partial x} \right)^2 - \frac{\hbar i}{2m} \frac{\partial^2 W}{\partial x^2} + V(x) = E \quad (4.10)$$

or,

$$\frac{\hbar}{i} \frac{\partial^2 W}{\partial x^2} + \left(\frac{\partial W}{\partial x} \right)^2 = 2m(E - V(x)) \quad (4.11)$$

Classically, $2m(E - V(x)) = m^2 v^2 = p^2$. So call $2m(E - V(x)) = p_c^2$, where the sub script c denotes that it is from a classical analogue. Also call, $\frac{\partial W}{\partial x} = p(x, E)$. Hence,

$$\frac{\hbar}{i} \frac{\partial p(x, E)}{\partial x} + p^2(x, E) = p_c^2 \quad (4.12)$$

Note that if $\hbar \rightarrow 0$, then first term in the LHS vanish and we get,

$$p(x, E) \rightarrow p_c^2$$

We call $p(x, E)$ as the Quantum Momentum Function, or QMF.

We also define the quantum action variable just like in classical mechanics as,

$$J(E) = \oint_C p(x, E) dx \quad (4.13)$$

C encloses all the turning points(the points at which $p_c^2 = 0$).

5 Equivalence Between Schrodinger Equation

We have actually started from Schrodinger equation to arrive at the Quantum Hamilton Jacobi equation. Here, we can verify this again. Note that,

the probability amplitude of going from one initial state $|x_1\rangle$ to a final state $|X_2\rangle$ is,

$$\langle x_1|X_2\rangle = e^{\frac{i}{\hbar}S} = e^{\frac{i}{\hbar}W} e^{-\frac{i}{\hbar}Et}$$

This is precisely the wave function. So, the time independent part of the wave function is,

$$\psi(x, E) = e^{\frac{i}{\hbar}W(x,E)}$$

Note that,

$$\frac{\partial\psi}{\partial x} = e^{\frac{i}{\hbar}W(x,E)} \frac{\partial W}{\partial x} \frac{i}{\hbar}$$

Hence,

$$p(x, E) = \frac{\partial W}{\partial x} = \frac{\hbar}{i} \frac{1}{\psi} \frac{\partial\psi}{\partial x} \quad (5.1)$$

Substitute the above in the Quantum Hamilton Jacobi equation(time independent), equation 4.11.

$$\begin{aligned} \frac{\hbar}{i} \frac{\partial}{\partial x} \left(\frac{\hbar}{i} \frac{1}{\psi} \frac{\partial\psi}{\partial x} \right) + \left(\frac{\hbar}{i} \frac{1}{\psi} \frac{\partial\psi}{\partial x} \right)^2 &= 2m(E - V(x)) \\ \Rightarrow -\hbar^2 \left(\frac{1}{\psi} \frac{\partial^2\psi}{\partial x^2} - \frac{1}{\psi^2} \left(\frac{\partial\psi}{\partial x} \right)^2 \right) - \hbar^2 \frac{1}{\psi^2} \left(\frac{\partial\psi}{\partial x} \right)^2 &= 2m(E - V(x)) \\ \Rightarrow \frac{\hbar^2}{2m} \frac{\partial^2\psi}{\partial x^2} + V(x)\psi &= E\psi \end{aligned}$$

No surprises here.

6 Quantization Rule

The turning points we find are like the boundaries of the system. Any wave function which is a solution to the problem we seek must be bounded between these turning points. Depending upon the energy level, the wave function will have different number of nodes. The nth energy level will correspond to n number of nodes. All these nodes must be between the turning points. Let x_0 be such a node of the wave function. So, we can write,

$$\psi = (x - x_0)\varphi(x)$$

$\varphi(x)$ do not have a zero at x_0 . Then using equation 5.1,

$$\begin{aligned} p(x, E) &= \frac{\hbar}{i} \frac{1}{(x - x_0)\varphi(x)} \left(\varphi(x) + (x - x_0) \frac{\partial\varphi(x)}{\partial x} \right) \\ &= \frac{\hbar}{i} \frac{1}{(x - x_0)} + \frac{\hbar}{i} \frac{1}{\varphi(x)} \frac{\partial\varphi(x)}{\partial x} \end{aligned}$$

This means that $p(x, E)$ has a pole at $x = x_0$, with a residue of $-i\hbar$. The position of these poles will depend on E and will change as E changes. So, these poles are called moving

poles. There can be another kinds of poles in ψ , due to singularities in the potential. These poles do not depend on energy level. Hence these are called fixed poles. As said before, the number of moving poles of $p(x, E)$ is equal to the energy level. Let the system be in n th energy level. Then $p(x, E)$ will have n first order poles between its turning points. Then, by residue theorem,

$$\oint_C p(x, E) dx = 2\pi i \sum_{j=1}^n (-i\hbar) = 2\pi n\hbar$$

Then using equation 4.13,

$$J(E) = 2\pi n\hbar \quad (6.1)$$

This is an exact quantization condition!

7 Example 1 : Simple Harmonic Oscillator

The harmonic oscillator potential is given by,

$$V(x) = \frac{1}{2}m\omega x^2 \quad (7.1)$$

Hence, equation 4.11 become,

$$\frac{\hbar}{i} \frac{\partial p(x, E)}{\partial x} + p^2(x, E) = 2m(E - \frac{1}{2}m\omega x^2) = p_c^2 \quad (7.2)$$

To find the turning points, put $p_c^2 = 0$.

$$\begin{aligned} \Rightarrow E - \frac{1}{2}m\omega x^2 &= 0 \\ \Rightarrow x_{1,2} &= \pm \sqrt{\frac{2E}{m\omega^2}} \end{aligned}$$

These are the turning points for harmonic oscillator. Every wave function which is a solution of this problem will be bounded between these two points. Hence, all the moving poles will be in between x_1 and x_2 . The fixed pole for this potential is when $x \rightarrow \infty$. So, Consider a contour, C which encloses both x_1 and x_2 . In the figure, Γ is another contour, whose radius $R \rightarrow \infty$. (see fig 1)

Clearly, $p(x, E)$ do not have any poles between the contours C and Γ . So using Cauchy's integral theorem,

$$\oint_{\Gamma} p(x, E) dx = \oint_C p(x, E) dx$$

Now, to simplify the calculations, we make a change of coordinates. put $x = \frac{1}{y}$. Now, as $x \in \Gamma$, we have,

$$x = Re^{i\theta} \quad (0 \leq \theta \leq 2\pi)$$

Then,

$$y = \frac{1}{R}e^{-i\theta}$$

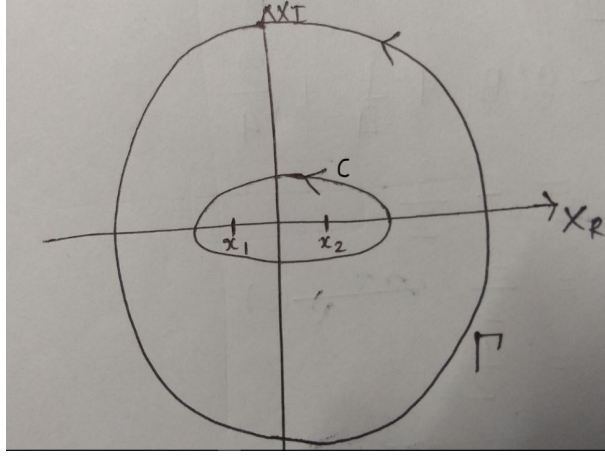


Figure 1: Turning points for SHO

Here, when we do the coordinate transformation, the direction of the contour reverses. As a result, the integral will have an overall $-ve$ sign. Also, $y \in \Gamma_0$, where Γ_0 is a contour with radius $r \rightarrow 0$ (as $R \rightarrow \infty$, $r = \frac{1}{R} \rightarrow 0$).

$$\begin{aligned} \oint_C p(x, E) dx &= - \oint_{\Gamma_0} p\left(\frac{1}{y}, E\right) d\left(\frac{1}{y}\right) \\ &= \oint_{\Gamma_0} \frac{\tilde{p}(y, E)}{y^2} dy \end{aligned}$$

Where $\tilde{p}(y, E) = p\left(\frac{1}{y}, E\right)$.

By this variable transformation, the quantum H-J equation 7.2 will become,

$$p^2\left(\frac{1}{y}, E\right) + \frac{\hbar}{i} \frac{\partial p\left(\frac{1}{y}, E\right)}{\partial(1/y)} = 2m \left(E - \frac{m\omega^2}{2y^2}\right)$$

Or,

$$\tilde{p}^2(y, E) + i\hbar y^2 \frac{\partial \tilde{p}(y, E)}{\partial y} = 2m \left(E - \frac{m\omega^2}{2y^2}\right) \quad (7.3)$$

We need to solve the integral $\oint_{\Gamma_0} \frac{\tilde{p}}{y^2} dy$. Γ_0 contains only one singularity, at $y = 0$. By residue theorem, we know that the only thing that will contribute to the integral is the residue of the integrand. For finding the residue and thereby finding the integral, we write $\tilde{p}(y, E)$ in Laurent series. Note that as $\hbar \rightarrow 0$, the function will approach $\tilde{p}_c(y, E)$. The RHS of the quantum H-J equation has a second order pole. So, if $\tilde{p}(y, E)$ has a pole which is of higher order than 1, then as the LHS of the quantum H-J equation will have a pole that has order more than 2, as LHS has a square of $\tilde{p}(y, E)$. So, $\tilde{p}(y, E)$ can have at most a first order pole. Hence,

$$\tilde{p}(y, E) = \sum_{n=0}^{\infty} a_n y^n + \frac{b_1}{y} \quad (7.4)$$

Hence,

$$\tilde{p}^2(y, E) = (a_0^2 + 2a_1b_1) + (2a_2b_1)y + (2a_0b_1)\frac{1}{y} + \frac{b_1^2}{y^2} + \dots$$

and,

$$y^2 \frac{\partial \tilde{p}(y, E)}{\partial y} = \sum_{n=0}^{\infty} n a_n y^{n+1} - b_1$$

Substitute these in equation 7.3,

$$(a_0^2 + 2a_1b_1) + (2a_2b_1)y + (2a_0b_1)\frac{1}{y} + \frac{b_1^2}{y^2} + \cdots + i\hbar \sum_{n=0}^{\infty} n a_n y^{n+1} - i\hbar b_1 = 2m \left(E - \frac{m\omega^2}{2y^2} \right)$$

Now compare the coefficients to get these equations,

$$\left. \begin{aligned} a_0^2 + 2a_1b_1 - i\hbar b_1 &= 2mE \\ 2a_2b_1 &= 0 \\ 2a_0b_1 &= 0 \\ b_1^2 &= -m^2\omega^2 \end{aligned} \right\} \quad (7.5)$$

Solving the last of the four equations give,

$$b_1 = \pm im\omega$$

Now, to remove the ambiguity in the sign, we can use,

$$\lim_{y \rightarrow 0} \tilde{p}^2(y, E) = \lim_{y \rightarrow 0} \left(2mE - \frac{m^2\omega^2}{y^2} \right) = -\frac{m^2\omega^2}{y^2} = \tilde{p}_c^2(y, E)$$

This imply,

$$\tilde{p}_c^2(y, E) = \pm \frac{im\omega}{y}$$

We use the convention of taking $\tilde{p}_c(y, E) = i|\tilde{p}_c(y, E)|$. This means $b_1 = im\omega$ is the choice (otherwise as $y \rightarrow 0$, $\tilde{p}(y, E) \rightarrow -\frac{im\omega}{y}$). Then the second and third equations will give us,

$$a_0 = 0 \quad a_2 = 0$$

From the first of those four equations,

$$\begin{aligned} (2a_1 - i\hbar)b_1 &= 2mE \\ \Rightarrow (2a_1 - i\hbar)(im\omega) &= 2mE \\ \Rightarrow a_1 &= \frac{E}{i\omega} + \frac{i\hbar}{2} \\ \Rightarrow a_1 &= \frac{2E - \hbar\omega}{2i\omega} \end{aligned}$$

The residue of the integrand is the coefficient of $\frac{1}{y}$ in $\frac{\tilde{p}}{y^2}$, which is a_1 . So by residue theorem,

$$\begin{aligned} \oint_{\Gamma_0} \frac{\tilde{p}(y, E)}{y^2} dy &= 2\pi i(a_1) \\ &= \frac{2\pi(2E - \hbar\omega)}{2\omega} \end{aligned}$$

Hence,

$$J(E) = \oint_C p(x, E) dx = \frac{2\pi(2E - \hbar\omega)}{2\omega} \quad (7.6)$$

Now, use the quantization condition given by equation 6.1,

$$2\pi n\hbar = \frac{2\pi(2E - \hbar\omega)}{2\omega}$$

Rearranging the terms,

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega \quad (7.7)$$

Which is indeed the energy spectrum of Simple Harmonic Oscillator!!

8 Example 2 : Harmonic Oscillator On A Half Line

The potential to our problem is,

$$V(x) = \begin{cases} \infty, & \text{if } x \leq 0. \\ \frac{1}{2}m\omega^2 x^2, & \text{if } x > 0. \end{cases} \quad (8.1)$$

The fixed poles in this problem are, $x = 0$ and $x \rightarrow \infty$. $x = 0$ also acts as a turning point of the problem. There is one more turning point for the problem, which is the same as in the last example.

$$x_2 = \sqrt{\frac{2E}{m\omega^2}}$$

$x_1 = -x_2$ was the other turning point in the previous problem. Note that if we do a parity inversion($x \rightarrow -x$), the system will have identical properties but the turning point changed from x_2 to x_1 . For quantization, we need to find out the integral,

$$J(E) = \oint_C p(x, E)dx$$

Where C is the contour that encloses all the moving poles between x_2 and 0. The diagram is shown in fig 2.

Now, by the symmetry,

$$\oint_C p(x, E)dx = \oint_{C'} p(x, E)dx = J(E)$$

Where C' is the contour that encloses all the moving poles between x_1 and 0 in the case of the flipped problem(as shown in the figure 2).

$p(x, E)$ will have one more pole at $x = 0$ due to the singularity in the potential. The contour γ_0 encloses this fixed pole. Now, we consider a contour, Γ , whose radius, $R \rightarrow \infty$. Then,

$$\begin{aligned} \oint_{\Gamma} p(x, E)dx &= \oint_{\gamma_0} p(x, E)dx + \oint_C p(x, E)dx + \oint_{C'} p(x, E)dx \\ &= 2J(E) + \oint_{\gamma_0} p(x, E)dx \end{aligned}$$

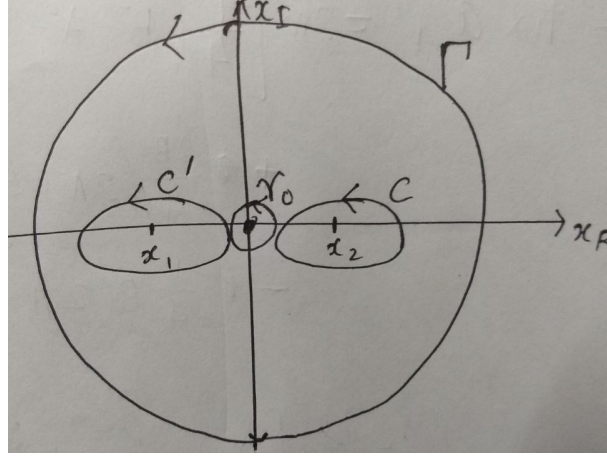


Figure 2: SHO on a half line

To find $\oint_{\gamma_0} p(x, E)dx$, we note that $p(x, E)$ has a singularity at $x = 0$, it will contribute a first order pole to the QMF. So, we can write the Laurent Series expansion of $p(x, E)$ around $x = 0$ as,

$$p(x, E) = \sum_{n=0}^{\infty} a_n x^n + \frac{b_1}{x} \quad (8.2)$$

Note that only b_1 , which is the residue of the integrand, will contribute to the integral. Plug this expansion in the quantum H-J equation to get,

$$\left(\frac{b_1^2}{x^2} + \dots \right) + \frac{\hbar}{i} \left(\frac{-b_1}{x^2} + \dots \right) = 2m \left(E - \frac{1}{2} m \omega^2 x^2 \right)$$

By comparing the coefficients of $\frac{1}{x^2}$,

$$b_1^2 + i\hbar b_1 = 0$$

Which has the solutions,

$$b_1 = 0, -i\hbar$$

$b_1 = 0$ is not an acceptable solution. If we put $b_1 = 0$ in the Laurent series, then the singularity at $x = 0$ is lost. So, the acceptable solution is $b_1 = -i\hbar$. Then by residue theorem,

$$\oint_{\gamma_0} p(x, E)dx = 2\pi i(-i\hbar) = 2\pi\hbar$$

Now, we need to find, $\oint_{\Gamma} p(x, E)dx$. For that, we use the change of variable, $x = \frac{1}{y}$. So the integral will become,

$$\oint_{\Gamma} p(x, E)dx = \oint_{\Gamma_0} \frac{\tilde{p}(y, E)}{y^2} dy$$

But this integral was exactly the same which we solved in the last example!!. Hence using equation 7.6,

$$\oint_{\Gamma} p(x, E)dx = \frac{2\pi(2E - \hbar\omega)}{2\omega} \quad (8.3)$$

We had,

$$\oint_{\Gamma} p(x, E) dx = 2J(E) + \oint_{\gamma_0} p(x, E) dx$$

And use the quantization rule equation 6.1,

$$\begin{aligned} \frac{2\pi(2E - \hbar\omega)}{2\omega} &= 2(2\pi n\hbar) + \hbar \\ \Rightarrow 2E &= 4n\hbar\omega + \hbar\omega + \frac{1}{2}\hbar\omega \end{aligned}$$

Rearranging terms,

$$E_n = \left((2n + 1) + \frac{1}{2} \right) \hbar\omega \quad (8.4)$$

9 Example 3 : The Eckart Potential

Using the results of Supersymmetric Quantum Mechanics and suitably adjusting the ground state energy to zero, we can write,

$$V(x) = w^2(x) - \frac{\hbar}{\sqrt{2m}} \frac{\partial w(x)}{\partial x} \quad (9.1)$$

Where,

$$w(x) = -A \coth \alpha x + \frac{B}{A} \quad (9.2)$$

$w(x)$ is called the superpotential for the problem. It is also known that for ground state ($E=0$),

$$w(x) = -\frac{\hbar}{\sqrt{2m}} \frac{1}{\psi_0} \frac{\partial \psi_0}{\partial x} \quad (9.3)$$

Using equation 5.1 for ground state,

$$p(x, E) = i\sqrt{2m} w(x) \quad (9.4)$$

The potential is,

$$\begin{aligned} V(x) &= A^2 \coth^2 \alpha x - 2B \coth \alpha x + \frac{B^2}{A^2} + \frac{\hbar\alpha A}{\sqrt{2m}} \operatorname{cosech}^2 \alpha x \\ &= A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right) \operatorname{cosech}^2 \alpha x + A^2 - 2B \coth \alpha x + \frac{B^2}{A^2} \\ &= A^2 + \frac{B^2}{A^2} + A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right) \operatorname{cosech}^2 \alpha x - 2B \coth \alpha x + \frac{B^2}{A^2} \end{aligned} \quad (9.5)$$

Hence the quantum H-J equation will look like,

$$p^2(x, E) + \frac{\hbar}{i} \frac{\partial p(x, E)}{\partial x} = 2m \left[E - A^2 - \frac{B^2}{A^2} - A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right) \operatorname{cosech}^2 \alpha x + 2B \coth \alpha x \right] \quad (9.6)$$

where x lies in the half plane ($x > 0$). Now to simplify things, we make a variable transformation,

$$y = e^{\alpha x}$$

Then,

$$\operatorname{cosech} \alpha x = \frac{1}{y - \frac{1}{y}} \quad \coth \alpha x = \frac{y + \frac{1}{y}}{y - \frac{1}{y}} \quad (9.7)$$

$$\frac{\partial}{\partial x} \rightarrow \frac{\partial}{\partial \left(\frac{\ln y}{\alpha}\right)} = y\alpha \frac{\partial}{\partial y}$$

$$dx \rightarrow d\frac{\ln y}{\alpha} = \frac{dy}{\alpha y}$$

$$p(x, E) = p\left(\frac{\ln y}{\alpha}, E\right) = \tilde{p}(y, E)$$

Then the H-J equation will become,

$$\tilde{p}^2(y, E) + \frac{\hbar\alpha y}{i} \frac{\partial \tilde{p}(y, E)}{\partial y} = 2m \left[E - A^2 - \frac{B^2}{A^2} - A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right) \frac{y^2}{(y^2 - 1)^2} + 2B \frac{y^2 + 1}{y^2 - 1} \right] \quad (9.8)$$

Equate the RHS to 0 to find the turning points. i.e,

$$E - A^2 - \frac{B^2}{A^2} - A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right) \frac{y^2}{(y^2 - 1)^2} + 2B \frac{y^2 + 1}{y^2 - 1} = 0$$

Or,

$$\left(E - A^2 - \frac{B^2}{A^2} \right) (y^2 - 1)^2 + 2B(y^2 + 1)(y^2 - 1) - A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right) y^2 = 0$$

The above equation is a degree 4 polynomial equation. This equation will have four solutions, out of which two are real and two are imaginary. y cannot take imaginary values because it is related to the position x via the variable transformation. So, we take only the real solutions (y_1 and y_2), one of which is positive, the other is negative. Now, parity inversion of the system will invert the problem to negative x axis. Apart from that, there will not be any changes to the system. From this symmetry, we can conclude that, $y_1 = -y_2$. So, to quantize the system, what we need to find is the integral,

$$J(E) = \oint_{C_1} \frac{\tilde{p}(y, E)}{\alpha y} dy \quad (9.9)$$

Where C_1 as shown below, encloses all the moving poles between $y = y_2$ and $y \rightarrow \infty$ ($x \rightarrow 0$). If C_2 is the contour that encloses all the poles in the flipped problem between $y = y_1$ and $y \rightarrow -\infty$.

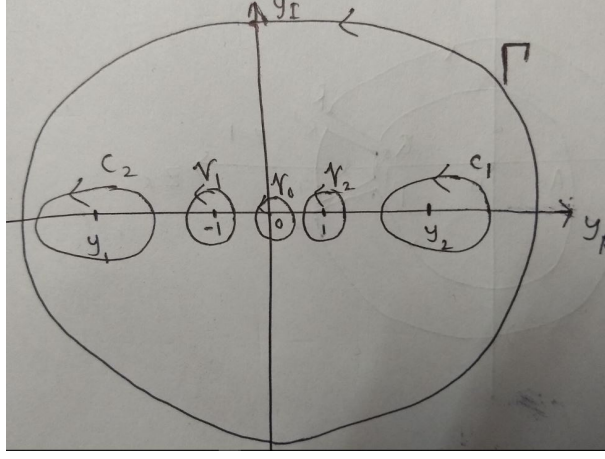


Figure 3: Contours for Eckart Potential

By symmetry,

$$J(E) = \oint_{C_1} \frac{\tilde{p}(y, E)}{\alpha y} dy = \oint_{C_2} \frac{\tilde{p}(y, E)}{\alpha y} dy \quad (9.10)$$

$\tilde{p}$ has poles at $y = \pm 1$. Hence, the integrand has poles at $y = 0, \pm 1$.

Consider a contour, Γ , whose radius $R \rightarrow \infty$. Let $\gamma_1, \gamma_0, \gamma_2$ be contours that encloses the poles at $-1, 0, 1$ respectively as shown in the figure 3. Then

$$\oint_{\Gamma} \frac{\tilde{p}(y, E)}{\alpha y} dy = \oint_{C_1} \frac{\tilde{p}(y, E)}{\alpha y} dy + \oint_{C_2} \frac{\tilde{p}(y, E)}{\alpha y} dy + \oint_{\gamma_0} \frac{\tilde{p}(y, E)}{\alpha y} dy + \oint_{\gamma_1} \frac{\tilde{p}(y, E)}{\alpha y} dy + \oint_{\gamma_2} \frac{\tilde{p}(y, E)}{\alpha y} dy$$

Or

$$\oint_{\Gamma} \frac{\tilde{p}(y, E)}{\alpha y} dy = 2J(E) + I_0 + I_1 + I_2 \quad (9.11)$$

Where,

$$I_0 = \oint_{\gamma_0} \frac{\tilde{p}(y, E)}{\alpha y} dy \quad I_1 = \oint_{\gamma_1} \frac{\tilde{p}(y, E)}{\alpha y} dy \quad I_2 = \oint_{\gamma_2} \frac{\tilde{p}(y, E)}{\alpha y} dy$$

To find I_1 , we expand $\tilde{p}(y, E)$ as a Laurent series around $y = -1$. $\tilde{p}$ will have a simple pole at $y = -1$. Then,

$$\tilde{p}(y, E) = \frac{b_1}{y+1} + \sum_{n=0}^{\infty} a_n (y+1)^n \quad (9.12)$$

In the evaluation of the integral, we need only the residue of the integrand which is only b_1 .

One other thing that we will require is,

$$\frac{y^2}{(y^2-1)^2} = -\frac{1}{4(y+1)} + \frac{1}{4(y+1)^2} + \frac{1}{4(y-1)} + \frac{1}{4(y-1)^2} \quad (9.13)$$

Now, put the Laurent series expansion in the quantum H-J equation and the LHS will become,

$$\left(\frac{b_1^2}{(y+1)^2} + \dots \right) + \frac{\hbar \alpha (y+1-1)}{i} \left(-\frac{b_1}{(y+1)^2} + \dots \right)$$

Now, compare the coefficients of $\frac{1}{(y+1)^2}$ to get b_1 . Hence,

$$b_1^2 + \frac{\hbar\alpha}{i}b_1 = -2m \cdot \frac{1}{4}A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right)$$

Solve the quadratic equation,

$$\begin{aligned} b_1 &= \frac{1}{2} \left[i\hbar\alpha \pm \sqrt{(-\hbar^2\alpha^2) - 4 \cdot A^2 \cdot 2m + 4\sqrt{2m}\hbar A} \right] \\ &= \frac{1}{2} \left[i\hbar\alpha \pm \sqrt{(i\hbar\alpha - 2\sqrt{2m}A)^2} \right] \\ &= \frac{i\hbar\alpha \pm i(\hbar\alpha - 2\sqrt{2m}A)}{2} \end{aligned}$$

Now, to find I_2 , expand $\tilde{p}$ in Laurent series around $y = 1$.

$$\tilde{p}(y, E) = \frac{d_1}{y-1} + \sum_{n=0}^{\infty} c_n (y-1)^n \quad (9.14)$$

The only coefficient we need to evaluate the integral is d_1 . The HJ equation will become,

$$\left(\frac{d_1^2}{(y-1)^2} + \dots \right) + \frac{\hbar\alpha(y-1+1)}{i} \left(-\frac{d_1}{(y-1)^2} + \dots \right)$$

Now, compare the coefficients of $\frac{1}{(y-1)^2}$ to get d_1 . Hence,

$$d_1^2 - \frac{\hbar\alpha}{i}d_1 = -2m \cdot \frac{1}{4}A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right)$$

Which gives,

$$d_1 = \frac{-i\hbar\alpha \pm i(\hbar\alpha - 2\sqrt{2m}A)}{2}$$

Now, to find I_0 , we need to write the Laurent series for $\tilde{p}(y, E)$ around $y = 0$. $\tilde{p}(y, E)$ is analytic at $y = 0$. Hence,

$$\tilde{p}(y, E) = \sum_{n=0}^{\infty} \bar{a}_n^2 y^n \quad (9.15)$$

To find the integral, the only relevant coefficient is a_0 . And, as $y \rightarrow 0$, the RHS of the quantum H-J equation will become,

$$\lim_{y \rightarrow 0} \left\{ 2m \left[E - A^2 - \frac{B^2}{A^2} - A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right) \frac{y^2}{(y^2-1)^2} + 2B \frac{y^2+1}{y^2-1} \right] \right\} = 2m \left(E - A^2 - \frac{B^2}{A^2} - 2B \right)$$

Now, apply the Laurent series in the QHJ equation as $y \rightarrow 0$ and compare the constants on both the sides,

$$\bar{a}_0^2 = 2m \left(E - A^2 - \frac{B^2}{A^2} - 2B \right)$$

Now, to find $\oint_{\Gamma} \frac{\tilde{p}(y, E)}{\alpha y} dy$, we do one more variable transformation,

$$y = \frac{1}{z}$$

Then,

$$\begin{aligned}\oint_{\Gamma} \frac{\tilde{p}(y, E)}{\alpha y} dy &= \oint_{\Gamma} \frac{\tilde{p}(1/z, E)}{\alpha y} d\frac{1}{z} \\ &= \oint_{\Gamma_0} \frac{q(z, E)}{\alpha z} dz\end{aligned}$$

Where the contour Γ_0 encloses the only pole of the integrand at $z = 0$ ($y \rightarrow \infty$). Under this variable transformation, the RHS of the H-J equation will become,

$$\begin{aligned}2m \left[E - A^2 - \frac{B^2}{A^2} - A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right) \frac{1}{z^2(\frac{1}{z^2} - 1)^2} + 2B \frac{\frac{1}{z^2} + 1}{\frac{1}{z^2} - 1} \right] \\ = 2m \left[E - A^2 - \frac{B^2}{A^2} - A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right) \frac{z^2}{(1 - z^2)^2} + 2B \frac{1 + z^2}{1 - z^2} \right]\end{aligned}$$

Therefore, as $z \rightarrow 0$ the RHS of quantum H-J equation will become,

$$\lim_{z \rightarrow 0} \left\{ 2m \left[E - A^2 - \frac{B^2}{A^2} - A \left(A - \frac{\hbar\alpha}{\sqrt{2m}} \right) \frac{z^2}{(1 - z^2)^2} + 2B \frac{1 + z^2}{1 - z^2} \right] \right\} = 2m \left(E - A^2 - \frac{B^2}{A^2} + 2B \right)$$

Now, write $q(z, E)$ as a Laurent series near $z = 0$. As $q(z, E)$ is analytic near 0, then,

$$q(z, E) = \sum_{n=0}^{\infty} \bar{b}_n^2 z^n \quad (9.16)$$

We only need the term $\bar{b}_0$ to evaluate the integral. For that, plug the series in the Quantum H-J equation for z variable and equate the constants. On LHS, the only constant term is $\bar{b}_0^2$. Hence,

$$\bar{b}_0^2 = 2m \left(E - A^2 - \frac{B^2}{A^2} + 2B \right)$$

Hence, we have,

$$\left. \begin{aligned} b_1 &= \frac{i\hbar\alpha \pm i(\hbar\alpha - 2\sqrt{2m}A)}{2} \\ d_1 &= \frac{-i\hbar\alpha \pm i(\hbar\alpha - 2\sqrt{2m}A)}{2} \\ \bar{a}_0^2 &= 2m \left(E - A^2 - \frac{B^2}{A^2} - 2B \right) \\ \bar{b}_0^2 &= 2m \left(E - A^2 - \frac{B^2}{A^2} + 2B \right) \end{aligned} \right\} \quad (9.17)$$

To fix the sign, we make help of equation 9.4. We look at the expression of $w(x)$ near the given poles as $E \rightarrow 0$. We have,

$$w = -A \coth \alpha x + \frac{B}{A} = -A \frac{y^2 + 1}{y^2 - 1} + \frac{B}{A} = -A \frac{1 + z^2}{1 - z^2} + \frac{B}{A} \quad (9.18)$$

Case 1 : near $y = -1$

$$w(y) \rightarrow -A \frac{1 + 1}{(y + 1)(-1 - 1)} + \frac{B}{A} = \frac{A}{y + 1} + \frac{B}{A}$$

and,

$$\tilde{p}(y, E) \rightarrow \frac{b_1}{y + 1} + a_0$$

Hence using 9.4, the correct sign we should take is the negative sign.

$$b_1 = i\sqrt{2m}A$$

Case 2 : near $y = 1$

$$w(y) \rightarrow -A \frac{1+1}{(1+1)(y-1)} \frac{B}{A} = -\frac{A}{y-1} + \frac{B}{A}$$

and,

$$\tilde{p}(y, E) \rightarrow \frac{d_1}{y-1}$$

Hence using 9.4, the correct sign we should take is the positive sign.

$$d_1 = -i\sqrt{2m}A$$

Case 3 : near $y = 0$

$$w(y) \rightarrow -A \frac{0+1}{0-1} + \frac{B}{A} = A + \frac{B}{A}$$

and,

$$\tilde{p}(y, E) \rightarrow \bar{a}_0(E \rightarrow 0)$$

$$\begin{aligned} \lim_{E \rightarrow 0} \bar{a}_0^2 &= 23 \left(-A^2 - \frac{B^2}{A^2} - 2B \right) \\ &= -2m \left(A + \frac{B}{A} \right)^2 \end{aligned}$$

$$a_0 = \pm i\sqrt{2m} \left(A + \frac{B}{A} \right)$$

So using 9.4, we should choose the positive root for $\bar{a}_0$.

Case 4 : near $z = 0$

$$w(z) \rightarrow -A \frac{1+0}{1-0} + \frac{B}{A} = -A + \frac{B}{A}$$

and,

$$q(z, E) \rightarrow \bar{b}_0(E \rightarrow 0)$$

$$\begin{aligned} \lim_{E \rightarrow 0} \bar{b}_0^2 &= 23 \left(-A^2 - \frac{B^2}{A^2} + 2B \right) \\ &= -2m \left(A - \frac{B}{A} \right)^2 \end{aligned}$$

$$\bar{a}_0 = \pm i\sqrt{2m} \left(A - \frac{B}{A} \right)$$

So using 9.4, we should choose the negative root for $\bar{a}_0$.

Hence, we have,

$$b_1 = iA\sqrt{2m} \quad (9.19)$$

$$d_1 = -iA\sqrt{2m} \quad (9.20)$$

$$\bar{a}_0 = \sqrt{2m \left(E - A^2 - \frac{B^2}{A^2} - 2B \right)} \quad (9.21)$$

$$\bar{b}_0 = -\sqrt{2m \left(E - A^2 - \frac{B^2}{A^2} + 2B \right)} \quad (9.22)$$

Now,

$$\begin{aligned} I_1 &= \lim_{y \rightarrow 1} \oint_{\gamma_1} \frac{\tilde{p}(y, E)}{\alpha y} dy = 2\pi i d_1 \\ &= \frac{2\pi}{\alpha} A\sqrt{2m} \end{aligned} \quad (9.23)$$

$$\begin{aligned} I_2 &= \lim_{y \rightarrow -1} \oint_{\gamma_2} \frac{\tilde{p}(y, E)}{\alpha y} dy = 2\pi i b_1 \\ &= \frac{2\pi}{\alpha} A\sqrt{2m} \end{aligned} \quad (9.24)$$

$$\begin{aligned} I_0 &= \lim_{y \rightarrow 0} \oint_{\gamma_0} \frac{\tilde{p}(y, E)}{\alpha y} dy = 2\pi i \bar{a}_0 \\ &= \frac{2\pi}{\alpha} i \sqrt{2m \left(E - A^2 - \frac{B^2}{A^2} - 2B \right)} \end{aligned} \quad (9.25)$$

$$\begin{aligned} I_\Gamma &= \lim_{z \rightarrow 0} \oint_{\Gamma_0} \frac{q(z, E)}{\alpha z} dy = 2\pi i \bar{b}_0 \\ &= -\frac{2\pi}{\alpha} i \sqrt{2m \left(E - A^2 - \frac{B^2}{A^2} + 2B \right)} \end{aligned} \quad (9.26)$$

$$(9.27)$$

Using the quantization rule, equation 6.1,

$$I_\Gamma - I_0 - I_1 - I_2 = 2 \cdot (2n\pi\hbar)$$

Hence, Which gives,

$$\begin{aligned} & -\frac{2\pi}{\alpha} i \sqrt{2m \left(E - A^2 - \frac{B^2}{A^2} + 2B \right)} - \frac{2\pi}{\alpha} i \sqrt{2m \left(E - A^2 - \frac{B^2}{A^2} - 2B \right)} - 2 \frac{2\pi}{\alpha} A\sqrt{2m} = 4n\pi\hbar \\ & \Rightarrow -i \left[\sqrt{2m \left(E - A^2 - \frac{B^2}{A^2} + 2B \right)} + \sqrt{2m \left(E - A^2 - \frac{B^2}{A^2} - 2B \right)} \right] = 2n\hbar\alpha + 2A\sqrt{2m} \\ & \Rightarrow -i \left[\sqrt{E - A^2 - \frac{B^2}{A^2} + 2B} + \sqrt{E - A^2 - \frac{B^2}{A^2} - 2B} \right] = 2 \left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right) \end{aligned}$$

Squaring both sides,

$$\begin{aligned}
& - \left[\left(E - A^2 - \frac{B^2}{A^2} + 2B \right) + \left(E - A^2 - \frac{B^2}{A^2} - 2B \right) \right] \\
& \quad - \left[2\sqrt{\left(E - A^2 - \frac{B^2}{A^2} + 2B \right) \left(E - A^2 - \frac{B^2}{A^2} - 2B \right)} \right] = 4 \left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right)^2 \\
& \Rightarrow -2 \left(E - A^2 - \frac{B^2}{A^2} \right) - 2\sqrt{\left(E - A^2 - \frac{B^2}{A^2} \right)^2 - 4B^2} = 4 \left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right)^2 \\
& \Rightarrow -\sqrt{\left(E - A^2 - \frac{B^2}{A^2} \right)^2 - 4B^2} = 2 \left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right)^2 + \left(E - A^2 - \frac{B^2}{A^2} \right)
\end{aligned}$$

Squaring once more,

$$\begin{aligned}
& \left(E - A^2 - \frac{B^2}{A^2} \right)^2 - 4B^2 = 4 \left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right)^4 + \left(E - A^2 - \frac{B^2}{A^2} \right)^2 + 4 \left(E - A^2 - \frac{B^2}{A^2} \right) \left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right)^2 \\
& \Rightarrow -B^2 - \left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right)^4 = \left(E - A^2 - \frac{B^2}{A^2} \right) \left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right)^2 \\
& \Rightarrow -\frac{B^2}{\left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right)^2} - \left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right)^2 = E - A^2 - \frac{B^2}{A^2}
\end{aligned}$$

Which gives us,

$$E_n = A^2 + \frac{B^2}{A^2} - \frac{B^2}{\left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right)^2} - \left(\frac{n\hbar\alpha}{\sqrt{2m}} + A \right)^2 \quad (9.28)$$

10 Example 4 : Particle In An Infinite Potential Well

The potential is given by,

$$V(x) = \begin{cases} 0, & \text{if } x < 0 < L. \\ \infty, & \text{if otherwise.} \end{cases} \quad (10.1)$$

Here, the turning points trivially are $x = 0$ and $x = L$. All the moving poles must be located in between these turning points. Let C be a rectangular contour enclosing all the moving poles in between the turning points. Then the Hamilton Jacobi equation in the region $0 < x < L$ is,

$$p^2(x, E) - i\hbar \frac{\partial p(x, E)}{\partial x} = 2mE \quad (10.2)$$

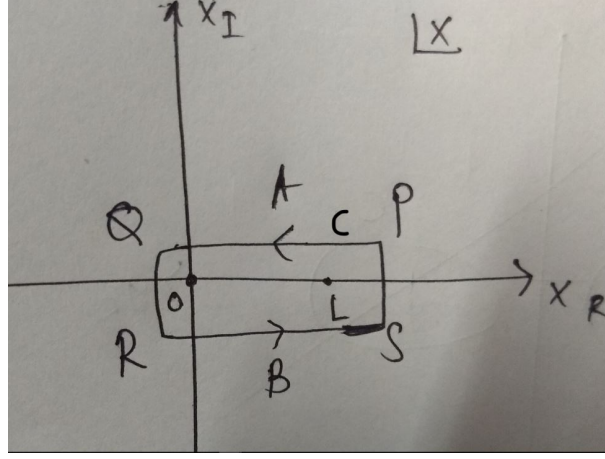


Figure 4: Potential Well

For simplifying the problem, we perform the substitution,

$$z = e^{\frac{2\pi i x}{L}} \quad (10.3)$$

Then the contour will be mapped to the contour PAQRBSP in the complex z plane. Corresponding to $x = 0$ and $x = L$, the pole in complex z plane is $z = 1$. The contour $P'Q'R'S'P'$ encloses the pole at $z = 1$.

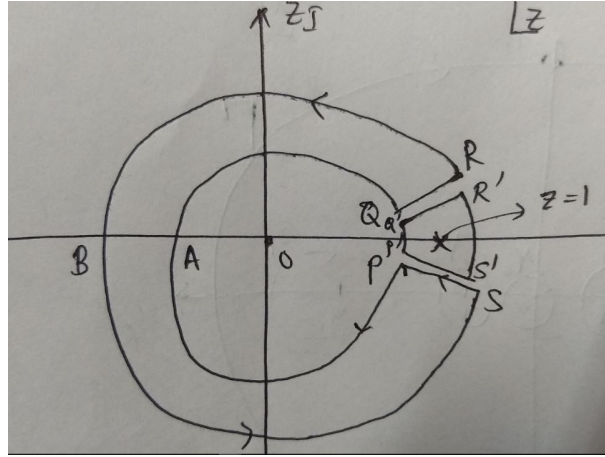


Figure 5: Potential Well after the substitution

Under the substitution,

$$p(x, E) = p\left(\frac{L \ln z}{2\pi i}, E\right) = \tilde{p}(z, E)$$

$$d\left(\frac{L \ln z}{2\pi i}\right) = \frac{L}{2\pi i} \cdot \frac{1}{z} dz$$

Hence, the action variable,

$$J(E) = \oint_C p(x, E) dx = \frac{L}{2\pi i} \oint_{C_1} \frac{\tilde{p}(z, E)}{z} dz \quad (10.4)$$

And the Quantum H-J equation become,

$$\tilde{p}^2(z, E) + \frac{2\pi\hbar z}{L} \frac{\partial \tilde{p}(z, E)}{\partial z} = 2mE \quad (10.5)$$

Consider three clockwise contours, Γ , γ and $\tau = P'Q'R'S'P'$.

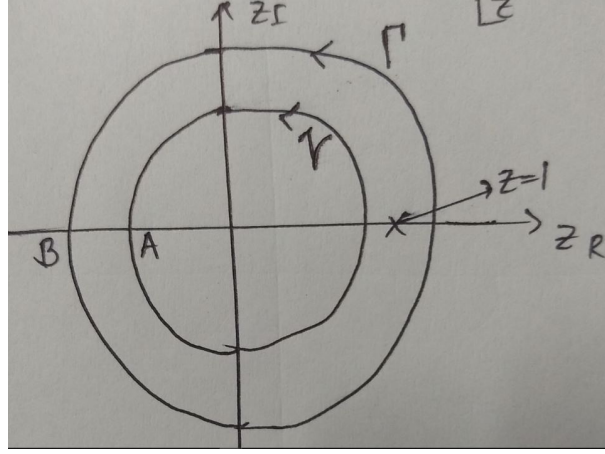


Figure 6:

Then,

$$J(E) = \frac{L}{2\pi i} \left[\oint_{\Gamma} \frac{\tilde{p}(z, E)}{z} dz - \oint_{\gamma} \frac{\tilde{p}(z, E)}{z} dz - \oint_{\tau} \frac{\tilde{p}(z, E)}{z} dz \right] \quad (10.6)$$

Calculation of Integral over Γ :

The contour encloses two poles, one the pole of $\tilde{p}$ at $z = 1$ and the other at $z = 0$ due to the denominator z . There are no poles outside the contour. So, Enlarge the contour to a very large size, and make a substitution,

$$z = \frac{1}{y}$$

Then as in the previous example,

$$\oint_{\Gamma} \frac{\tilde{p}(z, E)}{z} dz = \oint_{\Gamma'} \frac{\bar{p}(y, E)}{y} dy \quad (10.7)$$

Where $\bar{p}(y, E) = \tilde{p}(1/y, E)$ and Γ' is a contour enclosing the only pole of the new integral at $y = 0$. The H-J equation in the y variable is,

$$\bar{p}^2(y, E) - \frac{2\pi\hbar y}{L} \frac{\partial \bar{p}}{\partial y} = 2mE \quad (10.8)$$

Write,

$$\bar{p}(y, E) = \sum_{n=0}^{\infty} a_n y^n \quad (10.9)$$

Substitute in the H-J equation and equate the constants as done before,

$$a_0^2 = 2mE$$

Choose the negative sign. Then,

$$a_0 = -\sqrt{2mE}$$

Then by contour integration,

$$\oint_{\Gamma} \frac{\tilde{p}(z, E)}{z} dz = -2\pi i \sqrt{2mE} \quad (10.10)$$

Calculation of Integral over γ :

Write,

$$\tilde{p}(z, E) = \sum_{n=0}^{\infty} b_n z^n \quad (10.11)$$

And do as above. Then,

$$\begin{aligned} b_0 &= \sqrt{2mE} \\ \Rightarrow \oint_{\gamma} \frac{\tilde{p}(z, E)}{z} dz &= 2\pi i \sqrt{2mE} \end{aligned} \quad (10.12)$$

Calculation of Integral over τ :

Write,

$$\tilde{p}(z, E) = \sum_{n=0}^{\infty} c_n (z-1)^n + \frac{c_{-1}}{z-1} \quad (10.13)$$

Apply in H-J equation,

$$\frac{c_{-1}^2}{(z-1)^2} + \dots - \frac{2\pi\hbar}{L}(z-1+1) \left(\frac{c_{-1}}{(z-1)^2} + \dots \right) = 2mE$$

Equating the coefficients of $(z-1)^{-2}$,

$$c_{-1}^2 - \frac{2\pi\hbar}{L} c_{-1} = 0$$

The acceptable solution is,

$$c_{-1} = \frac{2\pi\hbar}{L}$$

Hence,

$$\oint_{\tau} \frac{\tilde{p}(z, E)}{z} dz = 2\pi i \frac{2\pi\hbar}{L} \quad (10.14)$$

Then using equation 10.6 and the quantization rule 6.1,

$$\frac{L}{2\pi i} \left[-2\pi i \sqrt{2mE} - 2\pi i \sqrt{2mE} - \frac{4\pi^2 i \hbar}{L} \right] = 2\pi n \hbar$$

$$\begin{aligned} L \left[-2\sqrt{2mE} - \frac{2\pi\hbar}{L} \right] &= 2\pi n \hbar \\ \Rightarrow -2L\sqrt{2mE} &= 2\pi\hbar(n+1) \end{aligned}$$

Squaring and rearranging,

$$E_n = \frac{\pi^2 \hbar^2}{2mL^2} (n+1)^2 \quad (10.15)$$

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